— **RETICLE** · DI2 SUMMER CORPS · FINAL PRESENTATION

# RETICLE

essential | harmonized score | advantageous

**R**ationale **E**ngine **T**o **I**nform **C**RISPR **L**ist **E**ntities — one calibrated lens over 2,157 published CRISPR screens.

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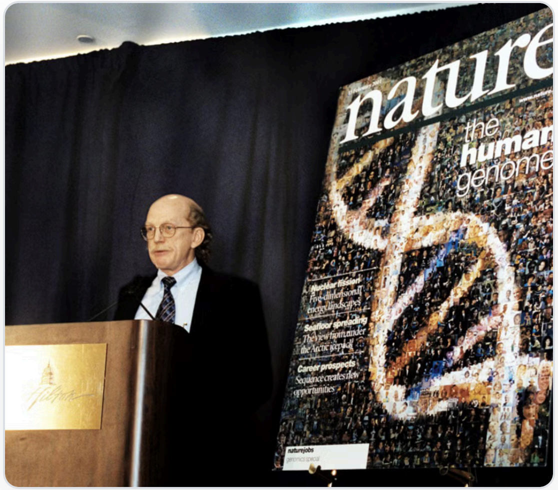
Washington University in St. Louis

00 — WHERE THIS STARTS

# A quarter of the human genome was finished *right here.*

## 00 — A SHORT HISTORY

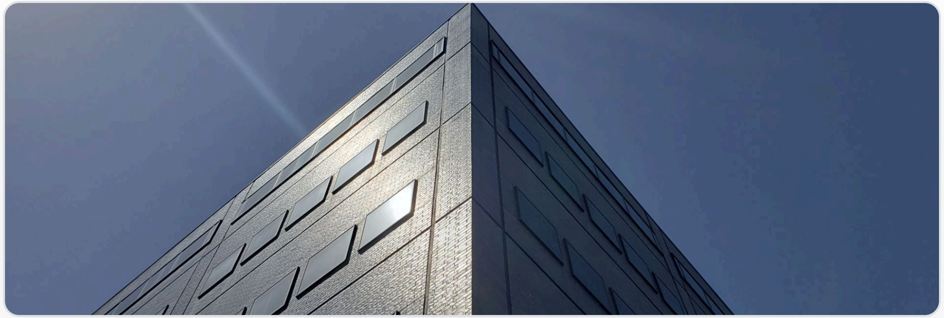
**Breaking ground** — the proclamation that started the work in St. Louis




**The announcement** — the finished sequence published in *Nature*, February 2001

| | | | |
|---|---|---|---|
| 2026 – | RETICLE | what they mean together | this project |
| 2003 – 2026 | CRISPR screens | what the parts do | 2,157 published · siloed |
| 1990 – 2003 | Human Genome Project | the parts list | WashU · 25% of the finished sequence |

# That sequence became the reference everything since is measured against.



**McDonnell Genome Institute** — where the work was done

**At the freezer** — checking a frozen sample pulled from cold storage, in cryo gloves

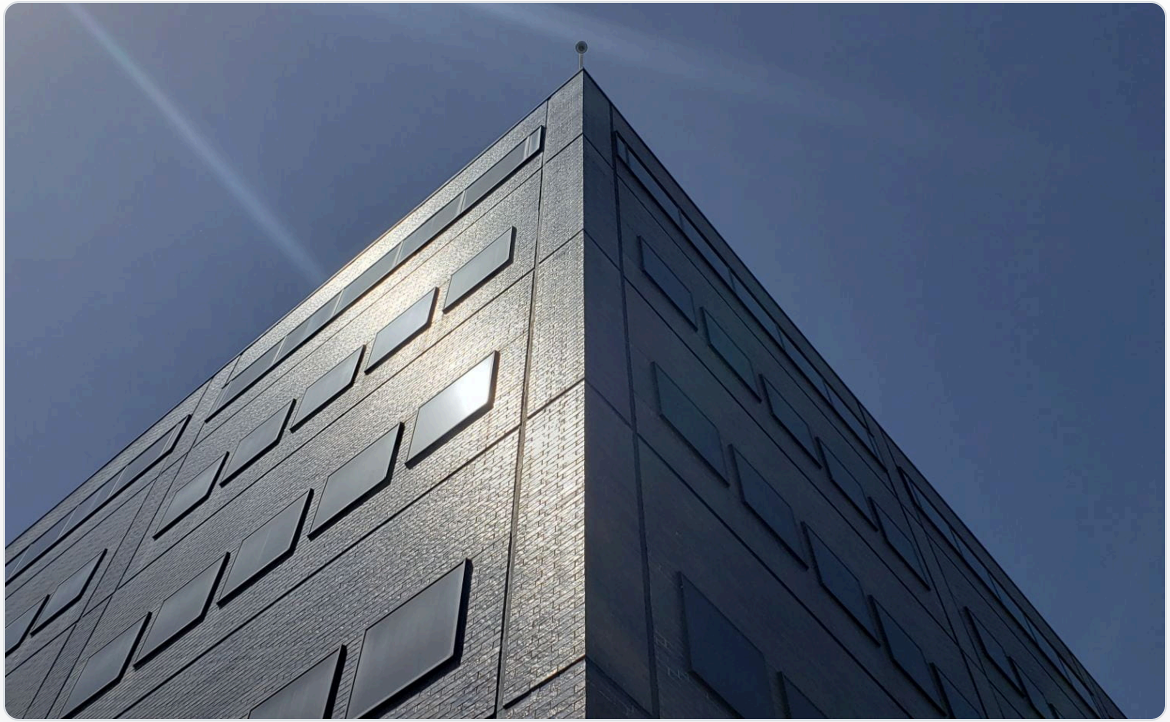
Every clinical genetic test compares a patient's DNA to it. Every disease-gene discovery since is anchored to it. Every CRISPR screen still maps its hits onto it — including **ours**.

The McDonnell Genome Institute was “a key player in the Human Genome Project, ultimately contributing **25 percent of the finished sequence**” — and more than 20% of the earlier draft.

Source: McDonnell Genome Institute, Washington University in St. Louis · genome.wustl.edu/about-2 · genome.wustl.edu/items/human-genome-project

# The lab that read the genome, and the lab trying to understand it.

THEN · 1990 – 2003



McDonnell Genome Institute — sequencing the parts list

NOW · 2026

Orvedahl Lab bench — or a RETICLE dashboard screenshot

`final_presentation_assets/orvedahl_lab.jpg`

Orvedahl Lab & RETICLE — working out what the parts *do*

Same campus, same problem, one layer further up. RETICLE is a **direct continuation** of the work that finished here.

# But we still don't know what most of the genome *does.*

**80%**

of protein-coding genes have a function that is **unknown or only partly understood** — two decades after the sequence was finished

first page / figure of the paper this number comes from

`final_presentation_assets/unknome_paper.png`

the published source for this figure — **[author, journal, year]**

00 — WHY THAT GAP MATTERS

# A gene nobody has characterised is a dead end for a real patient.

## Answers that never come

01 — the clinic

A genetic test finds a change in a gene no one has studied, so the result comes back as a **variant of unknown significance** — neither positive nor negative. The family waits.

## Drugs that never get made

02 — the pipeline

Drug discovery concentrates on proteins that are already well studied. The understudied majority is **never even evaluated** as a target — not because it failed, but because nobody looked.

## The biology in front of us

03 — this lab

The Orvedahl Lab studies cytokine-stress cell death in macrophages. The genes that matter most there are exactly the **buffered, under-studied** ones current methods are built to miss.

We'll come back to this at the end. For now — this is what the next few minutes are in service of.

# Sequencing gives you the parts list — not what the parts do.

a visual of the sequenced genome as a list — gene names running down the page,
a chromosome map, or a genome-browser view

`final_presentation_assets/genome_parts_list.png`

~20,000 protein-coding genes — every one located, named, and readable

- A parts list won't tell you which part the engine **can't run without** — for that, you have to remove one and see what breaks.

So we have every gene. **What does each one actually do?**

# A CRISPR screen switches off every gene — one per cell.

CRISPR mechanism diagram / animation still

`final_presentation_assets/crispr_mechanism.png`

Cas9 guided to one gene per cell — a library of guides covers the whole genome

THE EXPERIMENT, IN THREE STEPS

PERTURB

A guide library disables **each gene** — one per cell, across millions of cells

SELECT

The pool grows for weeks — on its own, or under a **drug or stress**

READ OUT

Sequence before and after — which knockouts **vanished**, which **took over**

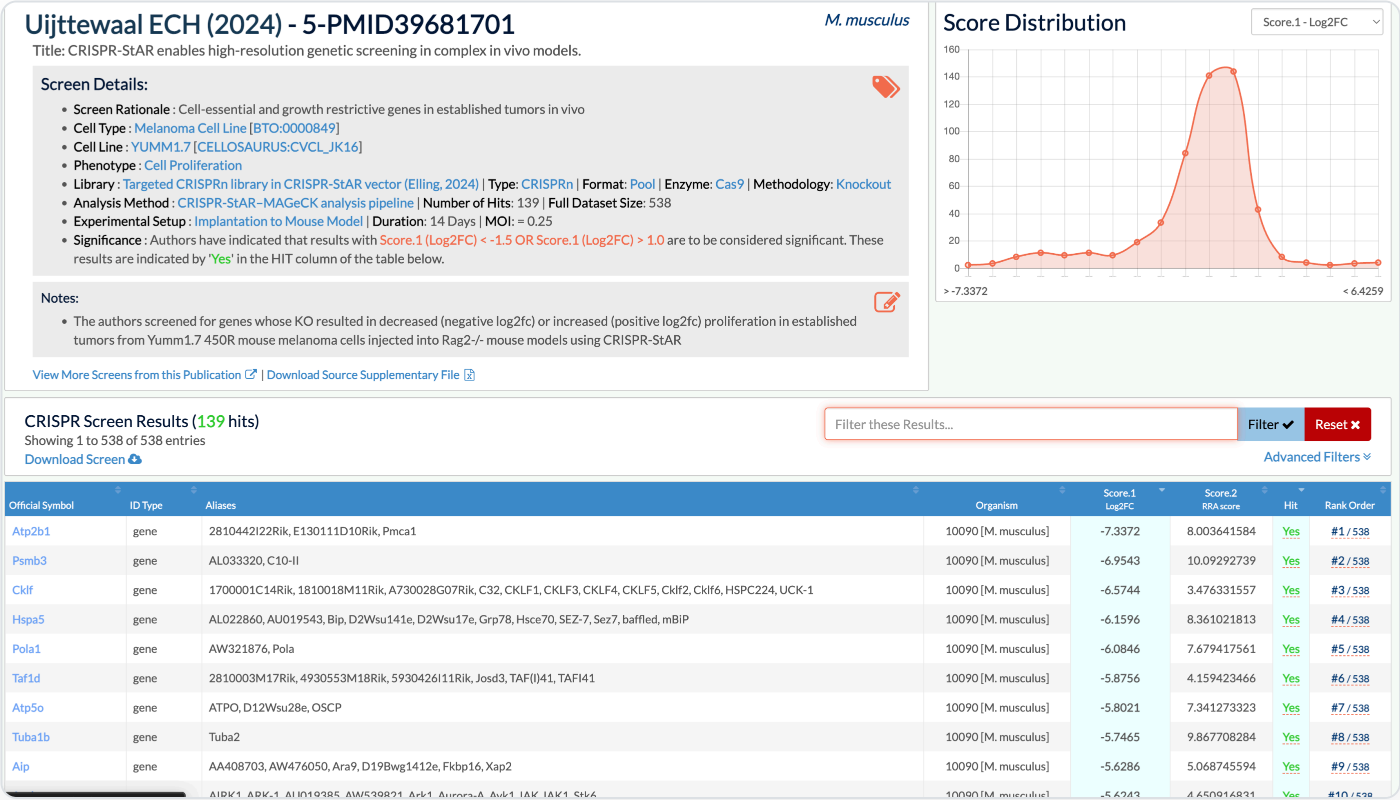
dropped out → the gene was essential

took over → losing it helped

Rare data — the measured result of actually **breaking** each gene. Every screen ends the same way: **one score per gene**. So what do we do with **thousands** of them?

# Today's tools each answer half the question.

THE PRIMARY EVIDENCE — THE SCREEN DATA ITSELF



## Uijttewaal ECH (2024) - 5-PMID39681701

*M. musculus*

Title: CRISPR-StAR enables high-resolution genetic screening in complex in vivo models.

Screen Details:

- Screen Rationale : Cell-essential and growth restrictive genes in established tumors in vivo
- Cell Type : Melanoma Cell Line [BTO:0000849]
- Cell Line : YUMM1.7 [CELLOSAURUS:CVCL_JK16]
- Phenotype : Cell Proliferation
- Library : Targeted CRISPRn library in CRISPR-StAR vector (Elling, 2024) | Type: CRISPRn | Format: Pool | Enzyme: Cas9 | Methodology: Knockout
- Analysis Method : CRISPR-StAR–MAGeCK analysis pipeline | Number of Hits: 139 | Full Dataset Size: 538
- Experimental Setup : Implantation to Mouse Model | Duration: 14 Days | MOI: = 0.25
- Significance : Authors have indicated that results with Score.1 (Log2FC) < -1.5 OR Score.1 (Log2FC) > 1.0 are to be considered significant. These results are indicated by 'Yes' in the HIT column of the table below.

Notes:

- The authors screened for genes whose KO resulted in decreased (negative log2fc) or increased (positive log2fc) proliferation in established tumors from Yumm1.7 450R mouse melanoma cells injected into Rag2-/- mouse models using CRISPR-StAR

View More Screens from this Publication | Download Source Supplementary File



CRISPR Screen Results (139 hits)
Showing 1 to 538 of 538 entries
Download Screen

Filter these Results... Filter Reset
Advanced Filters

| Official Symbol | ID Type | Aliases | Organism | Score.1 Log2FC | Score.2 RRA score | Hit | Rank Order |
|---|---|---|---|---|---|---|---|
| Atp2b1 | gene | 2810442I22Rik, E130111D10Rik, Pmca1 | 10090 [M. musculus] | -7.3372 | 8.003641584 | Yes | #1 / 538 |
| Psmb3 | gene | AL033320, C10-II | 10090 [M. musculus] | -6.9543 | 10.09292739 | Yes | #2 / 538 |
| Cklf | gene | 1700001C14Rik, 1810018M11Rik, A730028G07Rik, C32, CKLF1, CKLF3, CKLF4, CKLF5, Cklf2, Cklf6, HSPC224, UCK-1 | 10090 [M. musculus] | -6.5744 | 3.476331557 | Yes | #3 / 538 |
| Hspa5 | gene | AL022860, AU019543, Bip, D2Wsu141e, D2Wsu17e, Grp78, Hsce70, SEZ-7, Sez7, baffled, mBiP | 10090 [M. musculus] | -6.1596 | 8.361021813 | Yes | #4 / 538 |
| Pola1 | gene | AW321876, Pola | 10090 [M. musculus] | -6.0846 | 7.679417561 | Yes | #5 / 538 |
| Taf1d | gene | 2810003M17Rik, 4930553M18Rik, 5930426I11Rik, Josd3, TAF(I)41, TAFI41 | 10090 [M. musculus] | -5.8756 | 4.159423466 | Yes | #6 / 538 |
| Atp5o | gene | ATPO, D12Wsu28e, OSCP | 10090 [M. musculus] | -5.8021 | 7.341273323 | Yes | #7 / 538 |
| Tuba1b | gene | Tuba2 | 10090 [M. musculus] | -5.7465 | 9.867708284 | Yes | #8 / 538 |
| Aip | gene | AA408703, AW476050, Ara9, D19Bwg1412e, Fkbp16, Xap2 | 10090 [M. musculus] | -5.6286 | 5.068745594 | Yes | #9 / 538 |

**BioGRID ORCS** — the published screens. One gene, looked up across many experiments. **This is the evidence base RETICLE is built on** — and the only place the causal data lives.

SECOND TIER — ANNOTATION YOU CHECK AFTERWARDS

STRING screenshot

`final_presentation_assets/string_network.png`

**STRING** — *"is my hit already known to bind these partners?"*

GSEA / GO screenshot

`final_presentation_assets/gsea_pathway.png`

**GSEA · Gene Ontology** — *"do my hits pile up in one pathway?"*

The screens hold the answer. The annotation tools take **one list at a time** and know nothing about screens. So our output is screen data and published papers — **where does a researcher go from there?**

# Thousands of screens exist — and there is no way to use them together.

screen 1839 · IFN-γ · MAGeCK ← −

screen 0774 · TNF · log2FC → +

screen 1512 · IFN-γ · CERES ← −

screen 2043 · TNF · z-score ← −

screen 0918 · IFN-γ · CasTLE → +

One gene. Five published screens. Three call it **essential**, two call it **advantageous** — and nothing in the deposited data says which convention each one used.

01 · INSIDE ONE SCREEN

A hard cutoff throws away everything below the line · a gene that is absent is stored as if it scored **zero** · the stimulus, cell type and methods that give a score meaning never make it into the data at all

02 · BETWEEN SCREENS

Different stimuli, different cell lines, disagreeing directionality — **the same gene gets opposite signs** in two experiments and nothing reconciles them

03 · BEYOND THE SCREENS

Even once they line up, there is no way to ask the questions that matter: **co-essentiality** relationships, **dark-gene** candidates worth chasing, the published literature — and STRING, GO and the screens all live in separate tools

The result: an enormous body of causal evidence that is effectively **write-only** — deposited, and never reused together.

# This is where *RETICLE* comes in.

| | | | |
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The genome project let everyone stand on one lab's shoulders. **RETICLE does that for every published screen.**

# RETICLE

**R**ationale **E**ngine **T**o **I**nform **C**RISPR **L**ist **E**ntities

> *How does my screen compare to every screen in the CRISPR database?*

One question. Two ways in — **a whole screen**, or **a single gene**.

02 — SINCE THE MVP · OSEREMEN

# Multi-gene

Snapshot slides on what's new, then straight into the live demo — upload → compare → matched-screen drawer.

to be drafted · ~3:00

03 — SINCE THE MVP · AARON

# Single gene

One question — what does this gene do — and why it is still hard to answer.

03 — THE SEARCH

# You have a gene. You want to know what it does.

| Google? | Wikipedia? | ChatGPT? |
|---|---|---|

03 — THE SECOND PROBLEM

# Past studies did not treat every gene with the same attention.

2,000 genes — 9.7% the other **18,620**

Those 2,000 hold **52%** of every gene-to-paper link in the literature. The top 5,000 hold 74%.

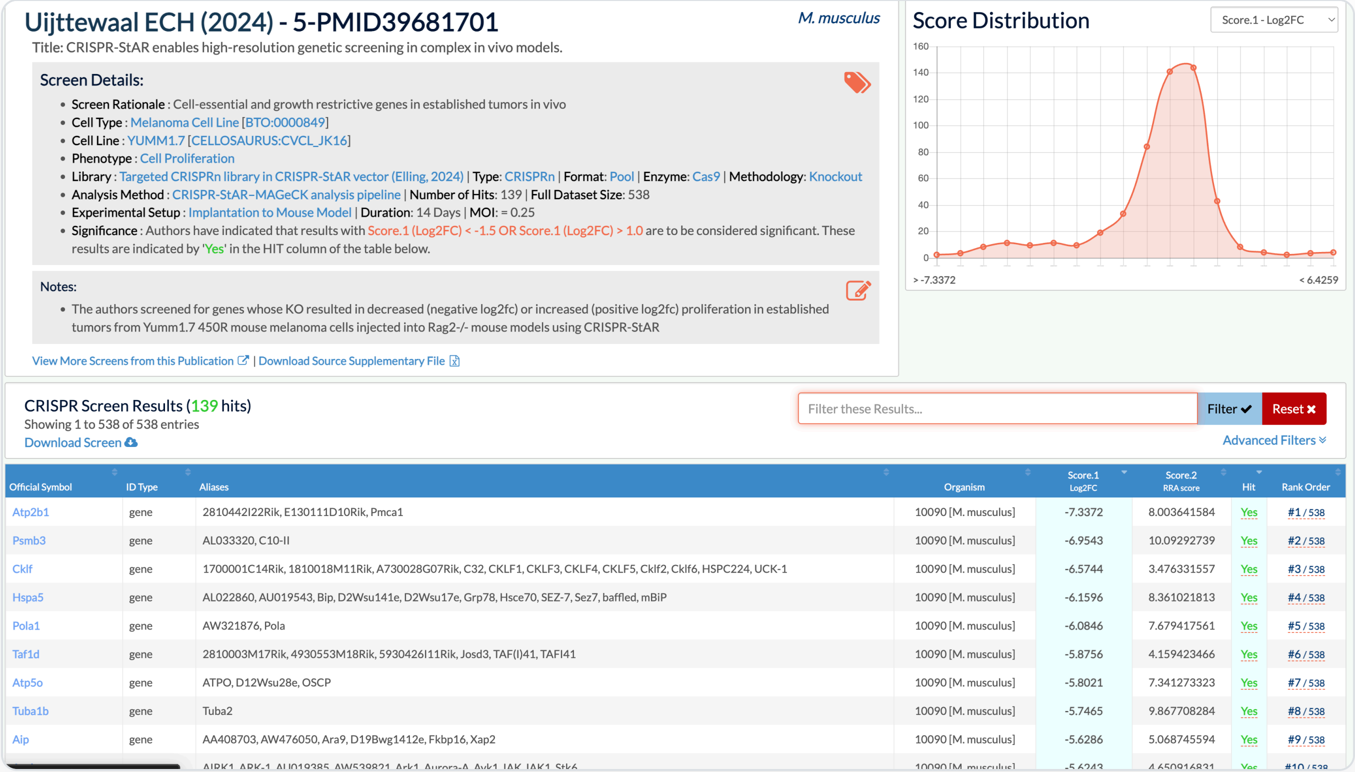
Attention was never evenly spread. **For most of the genome, the page would be blank.**

# RETICLE Wiki: ten trusted sources, assembled into one page.

| | | | | |
|---|---|---|---|---|
| NIH National Library of Medicine National Center for Biotechnology Information | UniProt | GENEONTOLOGY Unifying Biology | PubMed® | e!Ensembl |
| reactome | STRING | depmap portal | PDBe Protein Data Bank in Europe | BioGRID ORCS |

Ten sources, and the dark half is still dark.

# But a CRISPR screen never played favourites.




**Uijttewaal 2024** · 538 genes, 139 hits, one melanoma model — and every number a lab needs to say what it meant

- Screens like this one carry the evidence about **dark genes** — and 2,157 of them are published.
- Scored four different ways, they **cannot simply be read together**.

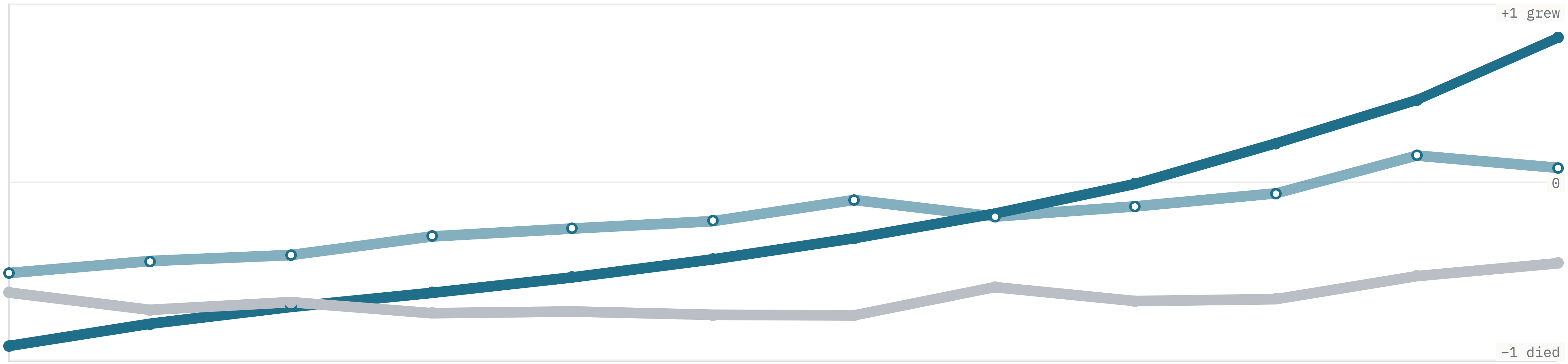
03 — WHAT WE BUILT

# RETICLE network

A map of gene-to-gene relationships we built ourselves, from CRISPR screening data alone.

03 — HOW IT IS BUILT

# Two genes are linked when they die in the same screens.




FANCD2 FANCA — tracks it, r = +0.35 RPL13 — essential everywhere, r = +0.11

1,335 screens measure all three. Sorted by FANCD2 and cut into twelve equal groups, each point is that group's mean. **An edge survives only if each gene is the other's best match** — which is what stops a gene that dies everywhere bonding with everything.

# Neighbours with the same job name the gene between them.

CTU2 28 papers

PREDICTED FUNCTION

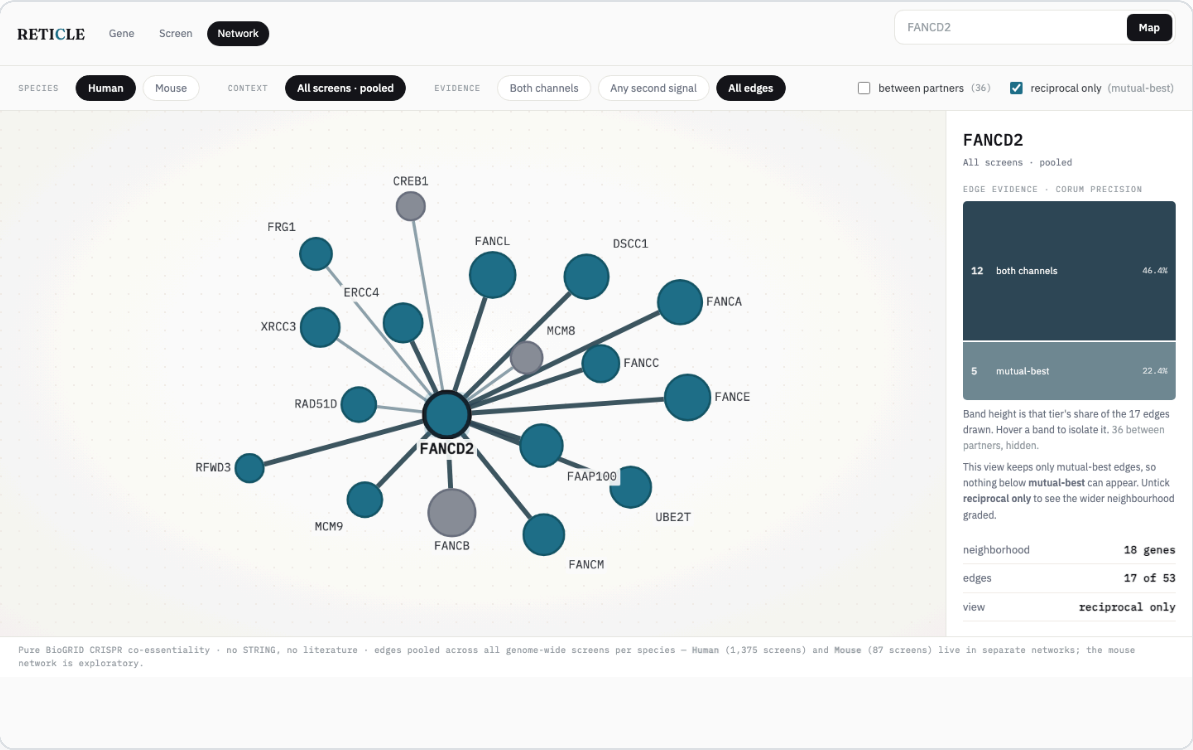
Those five are CTU2's mutual-best partners in the shipped network, at r = 0.26 to 0.32 over more than 1,300 screens — chosen by co-essentiality alone. **CTU2's own annotation says exactly this**, which is how we know the step works before trusting it on a gene where nobody can check.

# Our network rebuilt FANCD2's repair pathway.

FA CORE COMPLEX FANCG · FANCL · FANCE · FANCF · FANCM · UBE2T

ITS PARTNER FANCI

DOWNSTREAM REPAIR RAD51D · XRCC3 · MCM8 · MCM9 · RFWD3



04 — WHAT'S NEXT · AARON

# Future goals

Three things that would turn a working tool into a working method.

# *Wet lab verifications*

# *Connect with labs.*

# *Develop RETICLE API*

# Reflections

STUDENT CONTRIBUTOR

## Aaron Gao

My role was to build a data pipeline that transforms more than two thousand CRISPR-screen datasets into **interpretable insights about gene function**. I used Python, PostgreSQL, and WashU's RIS infrastructure to build the foundation, while learning to work with bioinformatics resources such as BioGRID ORCS, Gene Ontology, Reactome, and STRING. I also learned to apply LLMs to the interpretation layer, and — just as importantly — to **build the checks that verify what a model produces** against the underlying data before a user ever sees it. Another important skill I developed was cloud computing: with guidance from our technical lead, I learned how to turn a local application into a cloud-based tool other researchers could access. The most challenging part was that **real data does not arrive clean or consistent**, and much of the work was making it trustworthy before any analysis could mean anything. Working with researchers from different disciplines was one of the most valuable parts of this experience: I thought in pipelines and benchmarks, while the biology side of the team thought in pathways and whether a result would hold up at the bench. I learned that **a statistically strong result is not the same as a biologically meaningful one**, and that technical knowledge is most valuable when it addresses a real scientific need.

STUDENT CONTRIBUTOR

## Oseremen Ojiefoh

My role was on the front end — taking the team's research prototypes and turning them into **a web application a biologist could actually use**, and handling the deployment side so our work went live automatically. A lot of the job was **translation**: sitting between the science and the software, taking the researchers' questions and user stories and figuring out how to express them as an interface. The most challenging — and most rewarding — part was working across disciplines. I'm not a biologist, so I had to learn enough of the science to design for it, and the researchers had to see their ideas take shape as something clickable. I picked up real skills in cloud deployment, CI/CD, and single sign-on that I'd only read about before, and I learned **how much of good software is just clear communication** with people who think differently than you do.

06 — THANK YOU TO

# Our leaders

## Sourabh Sharma

Project Lead · DI2

## Saif Arif

Mini-DI2 Lead

06 — THANK YOU TO

# Our faculty

## Dr. Anthony Orvedahl

Principal Investigator · Orvedahl Lab

## Eli Perlin

Research Technician · Orvedahl Lab

06 — THE TEAM

# The team

## Oseremen Ojiefoh

BS Electrical Engineering ’28
web platform · deployment

## Aaron Gao

BS Computer Science & Statistics ’28
data pipeline · gene network

06 — THANK YOU TO

# Design support

## Tejas

Design Support

# Thank you.

| | | | |
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github.com/washu-dev/RETICLE